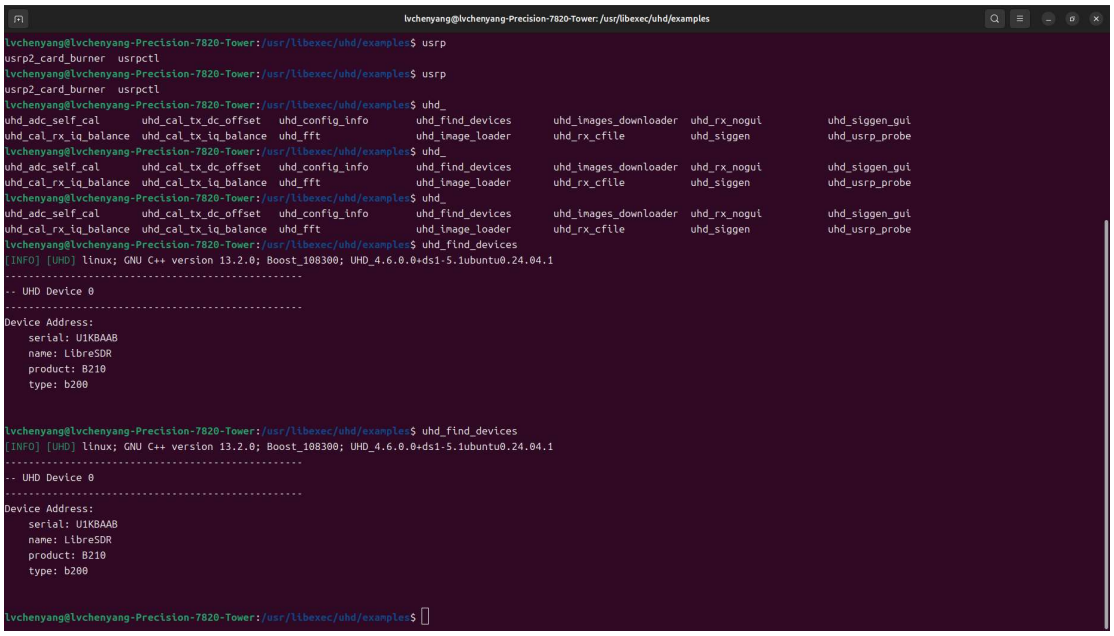
Quick Test of B210/B220mini

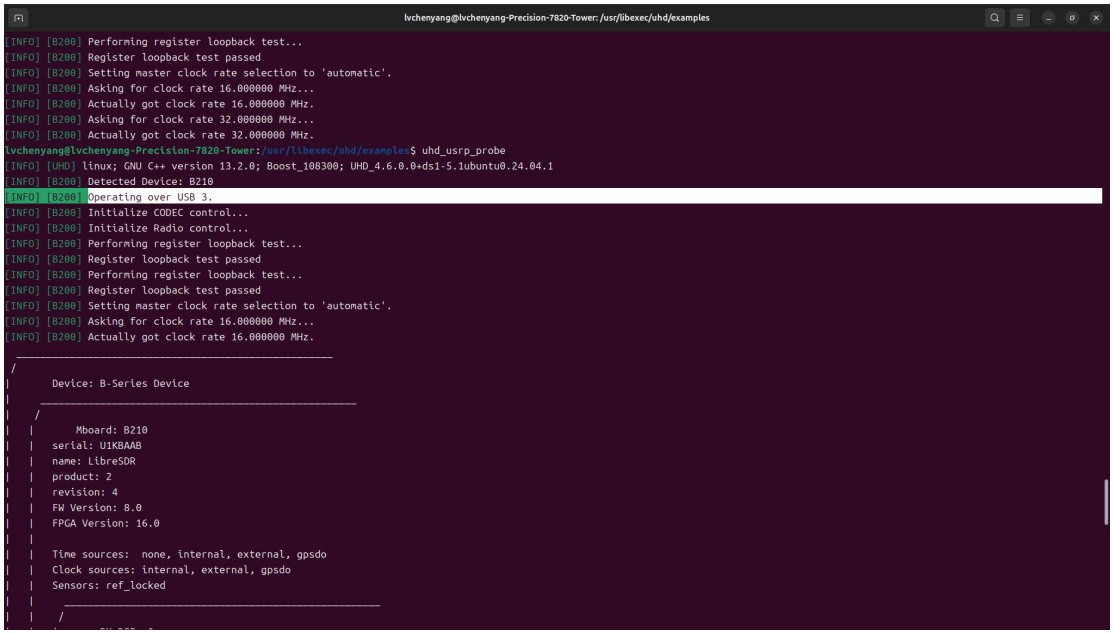
(It is also a production test process)

(but it will test power and sensitivity in combination with the signal source and the spectrum analyzer)

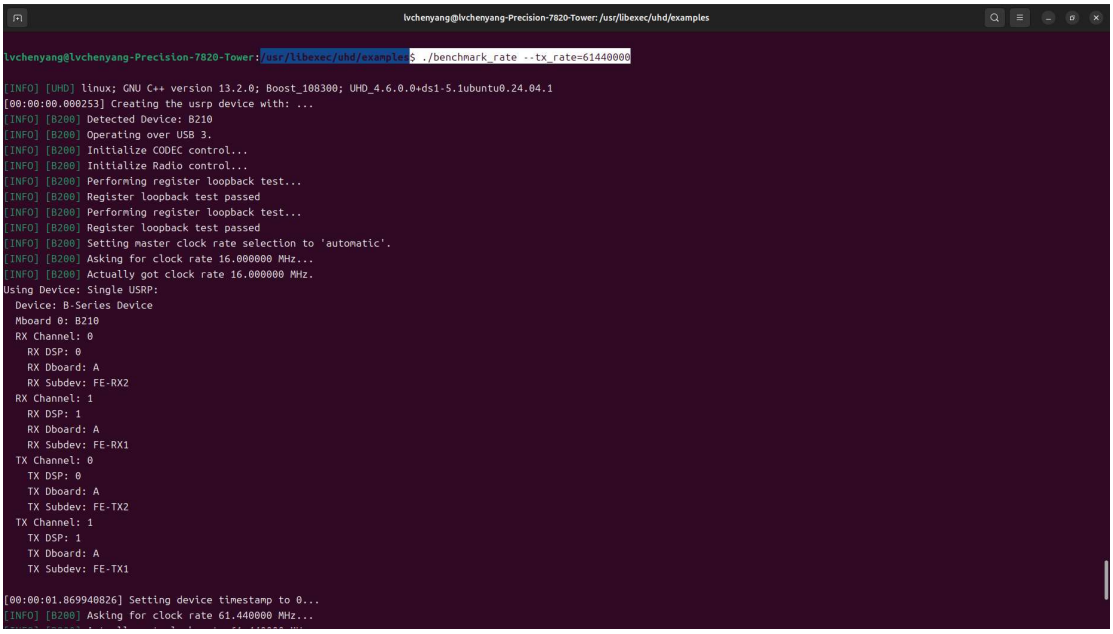
1. Currently, the FPGA bin file version only supports UHD4.1 and above
2. The FPGA files of B210mini and B220mini are not compatible
3. Uhd_find_devices



4. Uhd_usrp_probe Check whether it works in **USB 3.0** mode



5. /usr/libexec/uhd/examples$./benchmark_rate --tx_rate=61440000



```
lvchenyang@lvchenyang-Precision-7820-Tower: /usr/libexec/uhd/examples
    RX DSP: 1
    RX Dboard: A
    RX Subdev: FE-RX1
  TX Channel: 0
    TX DSP: 0
    TX Dboard: A
    TX Subdev: FE-TX2
  TX Channel: 1
    TX DSP: 1
    TX Dboard: A
    TX Subdev: FE-TX1

[00:00:01.869940826] Setting device timestamp to 0...
[INFO] [B200] Asking for clock rate 61.440000 MHz...
[INFO] [B200] Actually got clock rate 61.440000 MHz.
[WARNING] [AD936X] Selected Tx sample rate (61.44 MHz) is greater than
analog frontend filter bandwidth (56 MHz).
Setting TX spp to 2040
[00:00:02.235651534] Testing transmit rate 61.440000 Msps on 1 channels
[00:00:12.237121022] Benchmark complete.


Benchmark rate summary:
  Num received samples:     0
  Num dropped samples:      0
  Num overruns detected:    0
  Num transmitted samples:  614458200
  Num sequence errors (Tx): 0
  Num sequence errors (Rx): 0
  Num underruns detected:   0
  Num late commands:        0
  Num timeouts (Tx):        0
  Num timeouts (Rx):        0


Done!

lvchenyang@lvchenyang-Precision-7820-Tower:/usr/libexec/uhd/examples$
```

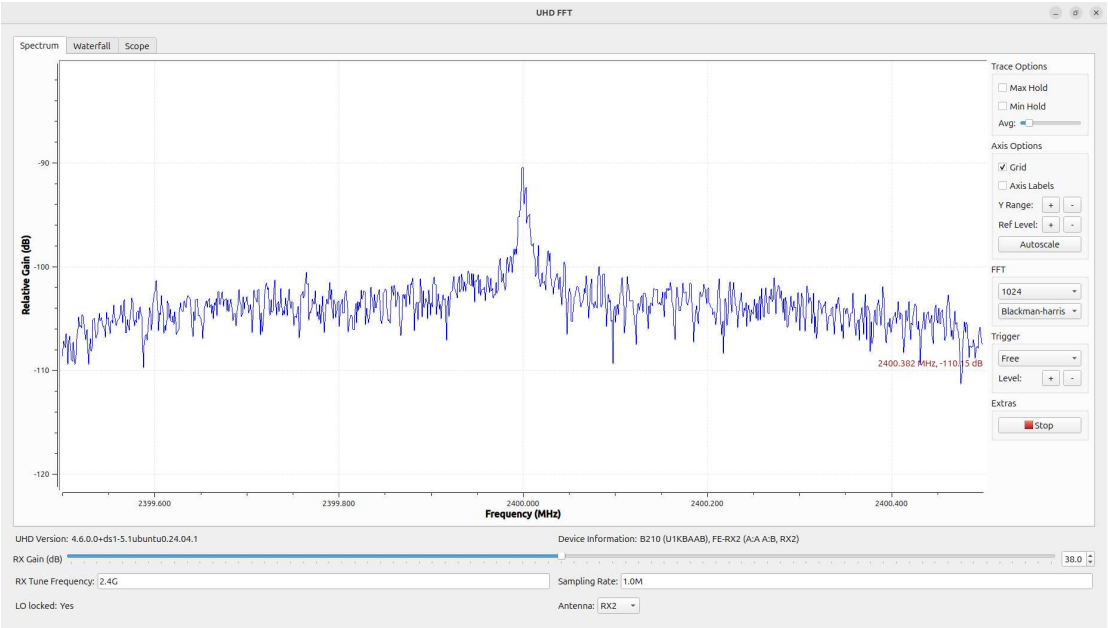
6. /usr/libexec/uhd/examples$ ./benchmark_rate --rx_rate=61440000

```
lvchenyang@lvchenyang-Precision-7820-Tower: /usr/libexec/uhd/examples
lvchenyang@lvchenyang-Precision-7820-Tower:/usr/libexec/uhd/examples$ ./benchmark_rate --rx_rate=61440000

[INFO] [UHD] linux; GNU C++ version 13.2.0; Boost_108300; UHD_4.6.0.0+ds1-5.1ubuntu0.24.04.1
[00:00:00.000170] Creating the usrp device with: ...
[INFO] [B200] Detected Device: B210
[INFO] [B200] Operating over USB 3.
[INFO] [B200] Initialize CODEC control...
[INFO] [B200] Initialize Radio control...
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Setting master clock rate selection to 'automatic'.
[INFO] [B200] Asking for clock rate 16.000000 MHz...
[INFO] [B200] Actually got clock rate 16.000000 MHz.
Using Device: Single USRP:
  Device: B-Series Device
  Mboard 0: B210
  RX Channel: 0
    RX DSP: 0
    RX Dboard: A
    RX Subdev: FE-RX2
  RX Channel: 1
    RX DSP: 1
    RX Dboard: A
    RX Subdev: FE-RX1
  TX Channel: 0
    TX DSP: 0
    TX Dboard: A
    TX Subdev: FE-TX2
  TX Channel: 1
    TX DSP: 1
    TX Dboard: A
    TX Subdev: FE-TX1

[00:00:01.768595763] Setting device timestamp to 0...
[INFO] [B200] Asking for clock rate 61.440000 MHz...
[INFO] [B200] Actually got clock rate 61.440000 MHz.
```

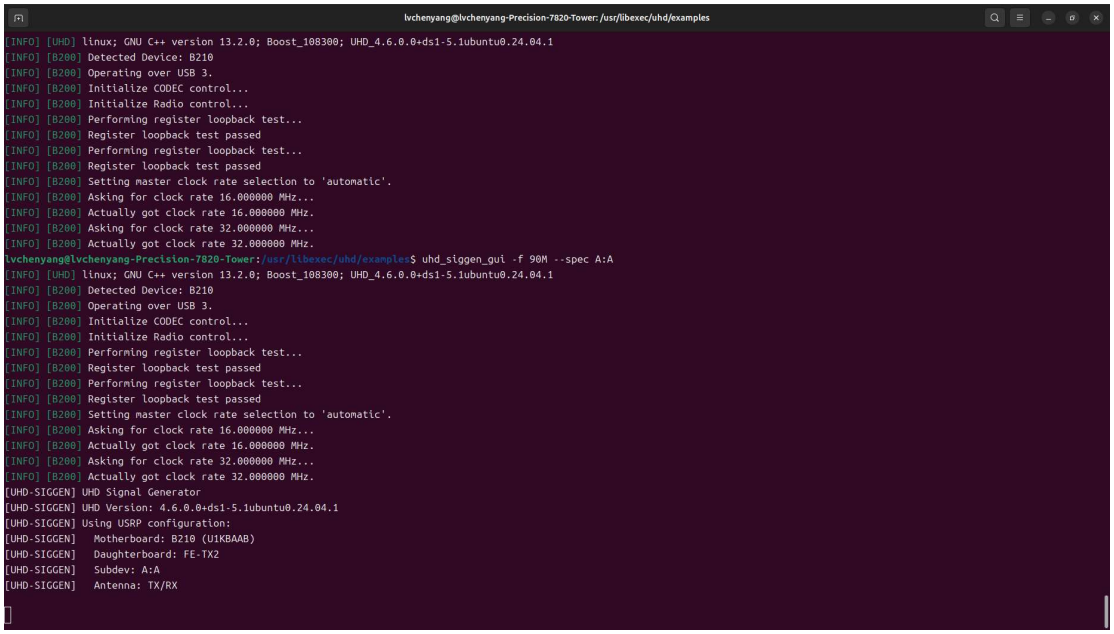
```
lvchenyang@lvchenyang-Precision-7820-Tower: /usr/libexec/uhd/examples
  RX Channel: 1
    RX DSP: 1
    RX Dboard: A
    RX Subdev: FE-RX1
  TX Channel: 0
    TX DSP: 0
    TX Dboard: A
    TX Subdev: FE-TX2
  TX Channel: 1
    TX DSP: 1
    TX Dboard: A
    TX Subdev: FE-TX1

[00:00:01.768595763] Setting device timestamp to 0...
[INFO] [B200] Asking for clock rate 61.440000 MHz...
[INFO] [B200] Actually got clock rate 61.440000 MHz.
[WARNING] [AD936X] Selected Rx sample rate (61.44 MHz) is greater than
analog frontend filter bandwidth (56 MHz).
[00:00:02.111508348] Testing receive rate 61.440000 Msps on 1 channels
[00:00:12.111896923] Benchmark complete.


Benchmark rate summary:
  Num received samples:     614390754
  Num dropped samples:      0
  Num overruns detected:    0
  Num transmitted samples:  0
  Num sequence errors (Tx): 0
  Num sequence errors (Rx): 0
  Num underruns detected:   0
  Num late commands:        0
  Num timeouts (Tx):        0
  Num timeouts (Rx):        0


Done!

lvchenyang@lvchenyang-Precision-7820-Tower:/usr/libexec/uhd/examples$
```

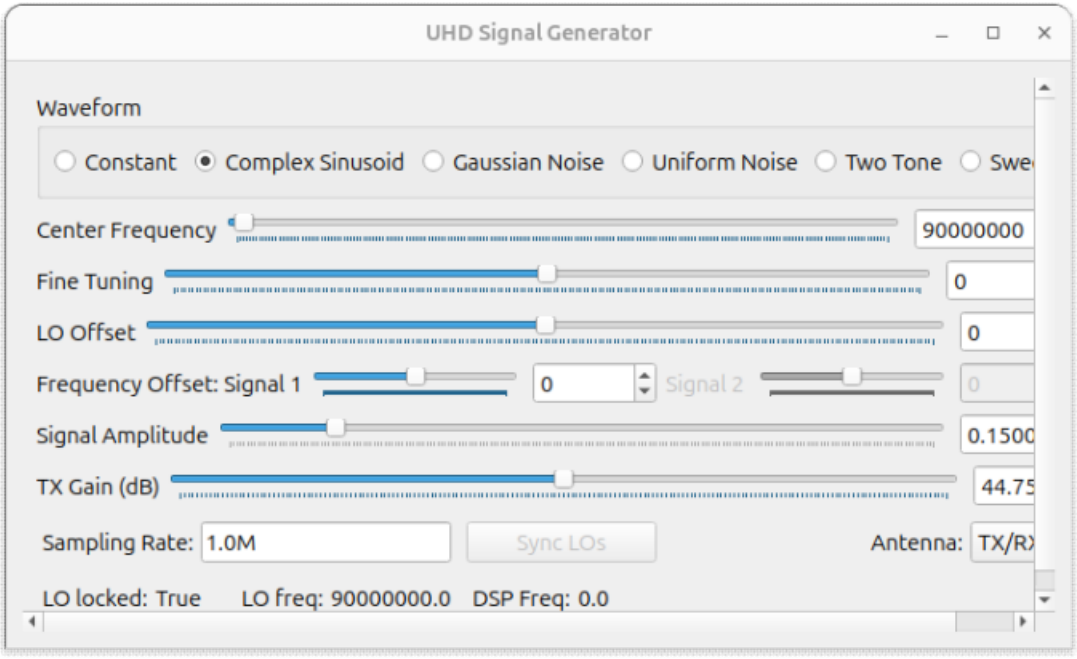
7. uhd_fft -f 2400M

```
lvchenyang@lvchenyang-Precision-7820-Tower: /usr/libexec/uhd/examples


Benchmark rate summary:
  Num received samples:     614390754
  Num dropped samples:      0
  Num overruns detected:    0
  Num transmitted samples:  0
  Num sequence errors (Tx): 0
  Num sequence errors (Rx): 0
  Num underruns detected:   0
  Num late commands:        0
  Num timeouts (Tx):        0
  Num timeouts (Rx):        0


Done!

lvchenyang@lvchenyang-Precision-7820-Tower:/usr/libexec/uhd/examples$ uhd_fft -f 2400M
[INFO] [UHD] linux; GNU C++ version 13.2.0; Boost_108300; UHD_4.6.0.0+ds1-5.1ubuntu0.24.04.1
[INFO] [B200] Detected Device: B210
[INFO] [B200] Operating over USB 3.
[INFO] [B200] Initialize CODEC control...
[INFO] [B200] Initialize Radio control...
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Setting master clock rate selection to 'automatic'.
[INFO] [B200] Asking for clock rate 16.000000 MHz...
[INFO] [B200] Actually got clock rate 16.000000 MHz.
[INFO] [B200] Asking for clock rate 32.000000 MHz...
[INFO] [B200] Actually got clock rate 32.000000 MHz.
lvchenyang@lvchenyang-Precision-7820-Tower:/usr/libexec/uhd/examples$ uhd_siggen_gui -f 90M --spec A:A
[INFO] [UHD] linux; GNU C++ version 13.2.0; Boost_108300; UHD_4.6.0.0+ds1-5.1ubuntu0.24.04.1
[INFO] [B200] Detected Device: B210
[INFO] [B200] Operating over USB 3.
[INFO] [B200] Initialize CODEC control...
```



8. uhd_siggen_gui --spec A:A




UHD Signal Generator
Waveform
Constant
Complex Sinusoid
Gaussian Noise
Uniform Noise
Two Tone
Center Frequency
90000000
Fine Tuning
0
LO Offset
0
Frequency Offset: Signal 1
0
Signal 2
0
Signal Amplitude
0.1500
TX Gain (dB)
44.75
Sampling Rate:
1.0M
Sync LOs
Antenna:
LO locked: True
LO freq: 90000000.0
DSP Freq: 0.0